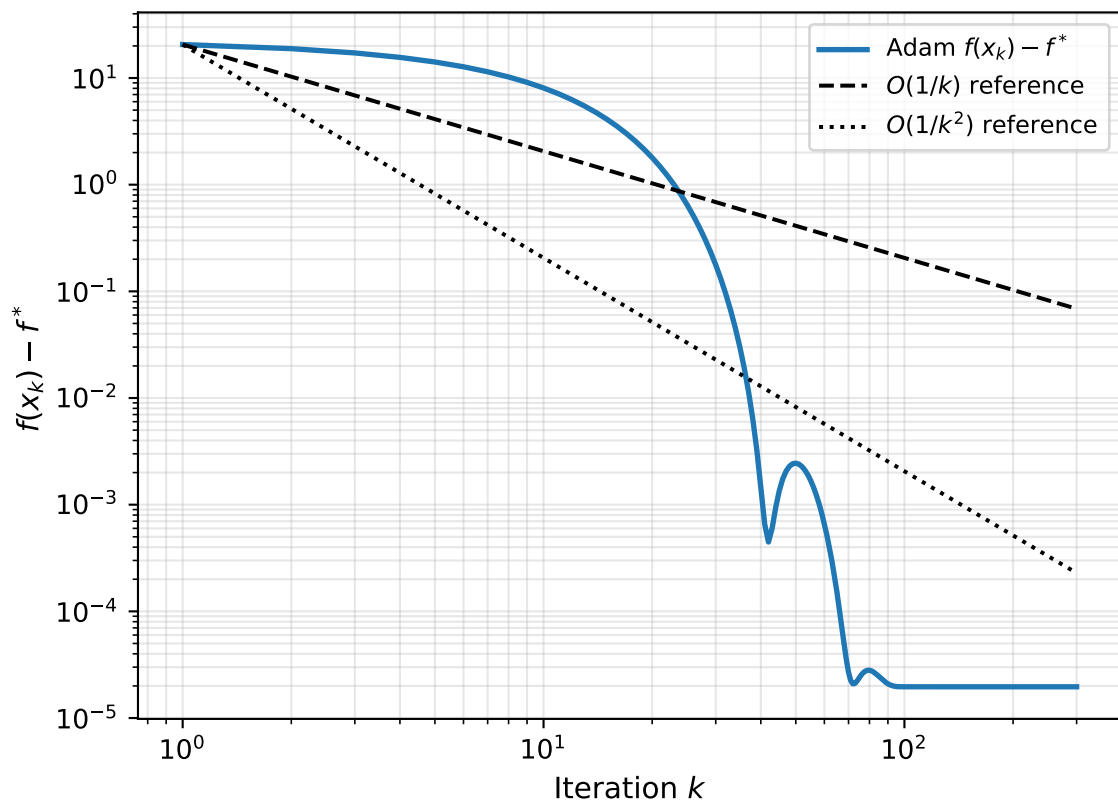


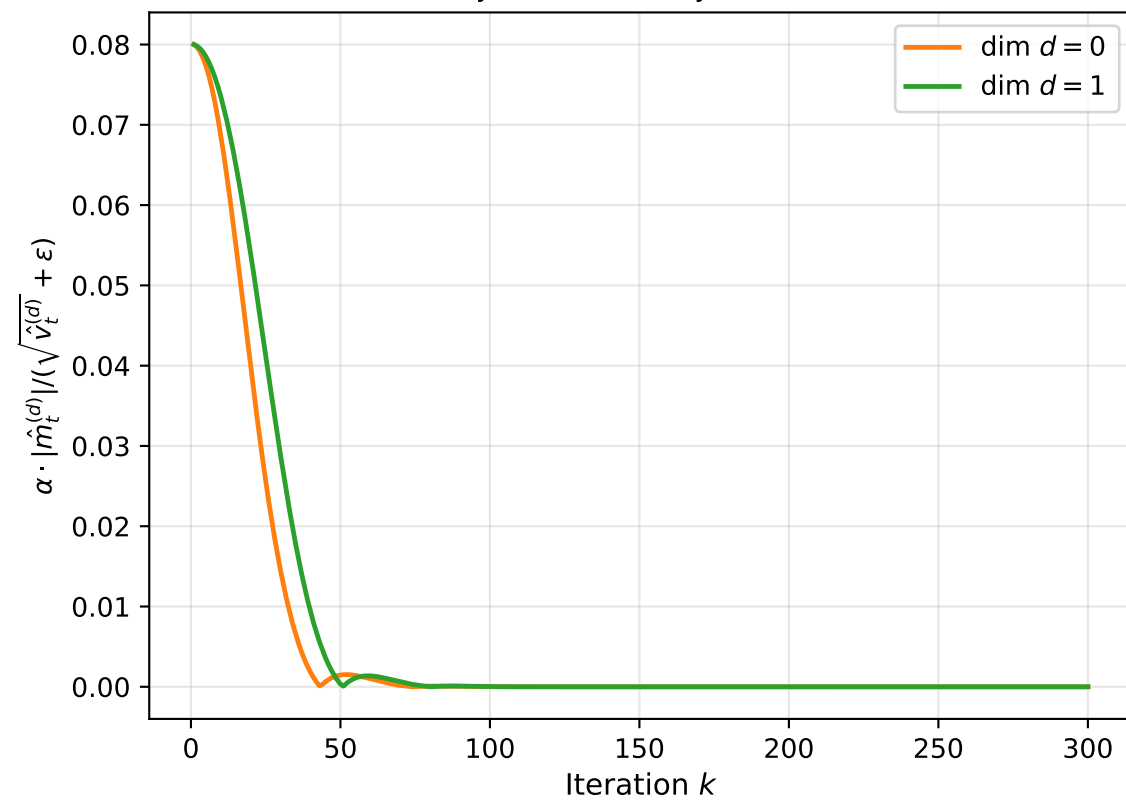
Adam Deep Analysis on Benchmark B

$$f(x) = (x_1 - 1)^2 + 5(x_2 - 2)^2 + \sin(x_1), \alpha = 0.08, \beta_1 = 0.82, \beta_2 = 0.999, x_0 = [-1, 4]$$

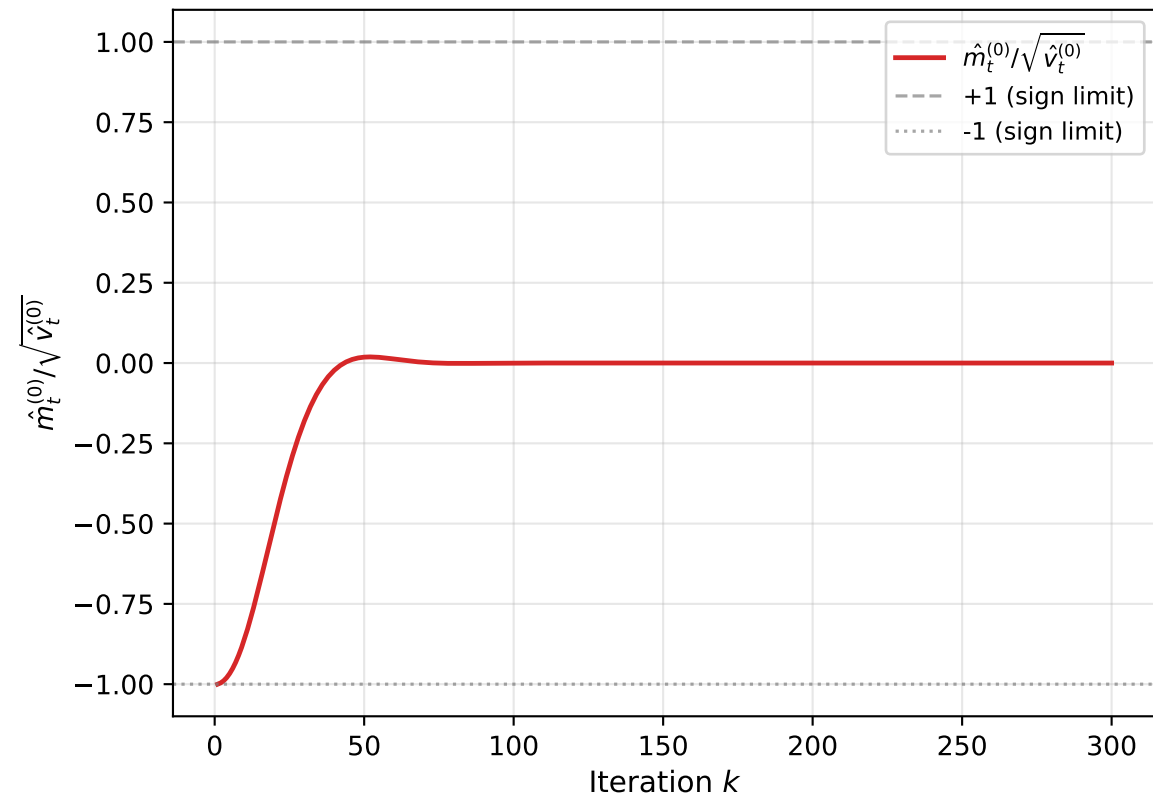
Suboptimality (log-log): Between $O(1/k)$ and $O(1/k^2)$



Effective Step Size per Dimension
(stays bounded away from zero)



Ratio $\hat{m}_t / \sqrt{\hat{v}_t}$ Approaches ± 1 (sign of gradient)



Bias Correction Decay: β_1^t Negligible after ~ 50 steps

